

Algebra 1
Unit 4
Lesson 9 Practice

Name _____
Date _____
Period _____

Use the following information to answer questions 1 through 5.

The lines of the equations $y - 7 = 3(x - 2)$ and $-\frac{1}{3}x + y = 1$ are shown on the same coordinate grid.

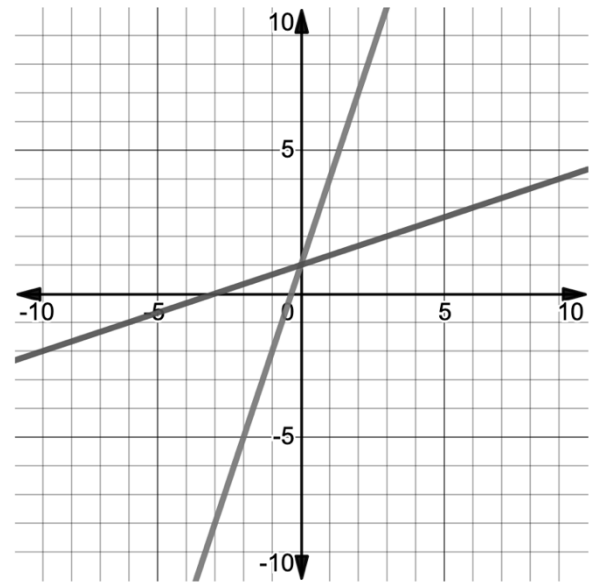
1. What is the slope of the line representing the equation $y - 7 = 3(x - 2)$?

2. What is the slope of the line representing the equation $-\frac{1}{3}x + y = 1$?

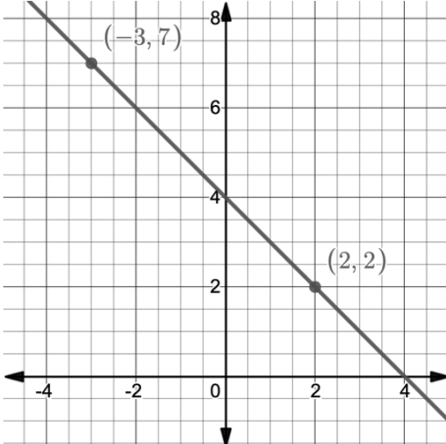
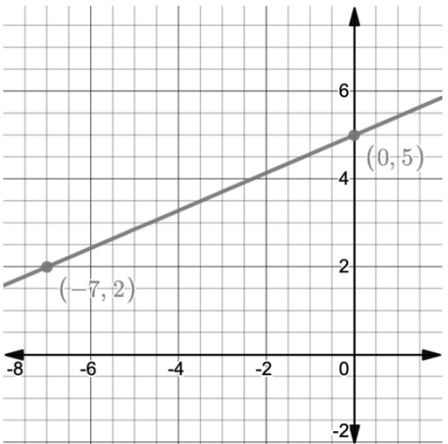
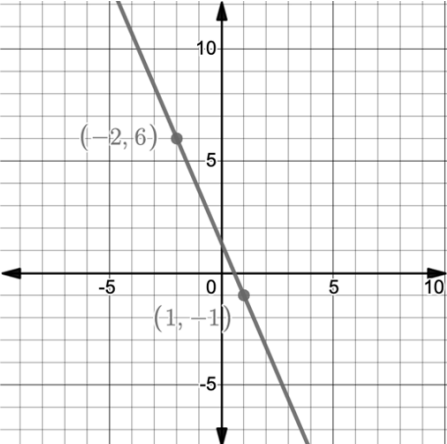
3. What is the slope of a line that is perpendicular to $-\frac{1}{3}x + y = 1$?

4. Explain how to determine if two lines are perpendicular.

5. Determine if the lines are parallel, perpendicular, or neither.



Use the following lines to answer questions 6 through 8.

Line D	Line G	Line K
		
Line M	Line N	Line P
$-x - y = 6$	$y = -x + 8$	$\frac{7}{3}x - y = -1$

6. Which lines are parallel?
7. Which lines are perpendicular?
8. What is the slope of a line that is parallel to Line P?
9. Determine if the lines of the two equations are parallel, perpendicular, or neither.
- | Equation H | Equation D |
|-----------------|-----------------|
| $-2x - 4y = -8$ | $-2x + 4y = -8$ |
10. Determine if the lines of the two equations are parallel, perpendicular, or neither.
- | Equation S | Equation W |
|----------------|--------------------|
| $-5x + y = -3$ | $y - 8 = 5(x - 3)$ |